

DURBAN, South AfricaWMO: **685880**Lat: **30.013S**Lon: **30.926E**Elev: **14**StdP: **101.16**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	10.4	1.1	4.1	15.6	3.1	4.7	15.5	11.8	18.4	10.5	19.0	1.5	320	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	30.2	23.9	29.2	23.5	28.6	23.2	25.3	28.5	24.7	27.8	24.3	27.3	6.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.1	27.1	23.9	18.7	26.9	23.2	17.9	26.4	77.6	28.8	75.3	28.0	73.3	27.3	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.9	9.8	8.8	6.5	34.2	1.4	1.7	5.5	35.4	4.7	36.4	3.9	37.3	2.8	38.6		
			4.5	26.2	1.1	0.6	3.7	26.6	3.0	27.0	2.4	27.3	1.6	27.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	24.3	24.8	24.0	21.9	19.6	17.5	16.9	18.2	19.2	20.2	21.6	23.2
	DBStd	3.26	1.91	1.81	1.87	2.00	1.87	1.69	1.69	1.86	2.15	2.19	2.15	2.06
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	143	0	0	0	1	9	35	49	24	15	7	2	0
	CDD10.0	3994	444	416	433	356	298	226	216	255	276	318	349	409
	CDD18.3	1094	186	182	174	107	48	10	6	21	42	67	101	151
	CDH23.3	5751	1181	1304	1065	470	195	78	51	78	97	167	324	741
(14) Wind	WSAvg	4.0	4.3	4.2	3.9	3.7	3.1	3.0	3.2	4.0	4.4	4.6	4.6	4.4
	PrecAvg	919	118	96	104	65	30	38	43	30	56	99	103	130
	PrecMax	1208	241	198	318	155	98	281	249	98	117	211	269	290
	PrecMin	444	68	25	28	10	2	0	0	2	10	27	42	46
	PrecStd	193	52	48	67	43	26	66	63	27	33	47	58	70
	DB	31.1	31.7	31.1	29.9	28.5	28.0	27.2	27.8	27.8	28.8	29.2	30.9	30.9
	MCWB	24.6	25.0	24.2	22.8	18.9	16.5	16.4	17.8	20.5	21.7	22.9	24.4	24.4
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	29.8	30.1	29.3	28.1	26.5	25.2	24.5	25.1	25.8	26.3	27.6	28.9	28.9
	2% MCWB	24.3	24.5	23.7	21.9	19.3	16.5	16.1	17.8	19.9	20.9	22.5	23.8	23.8
	5% DB	28.7	29.1	28.4	26.9	25.2	23.9	23.1	23.8	24.2	25.1	26.2	27.7	27.7
	5% MCWB	23.8	24.1	23.3	21.6	19.3	16.8	16.3	17.8	19.2	20.4	21.8	23.1	23.1
	10% DB	27.8	28.2	27.5	25.9	24.2	22.8	22.0	22.7	23.1	23.9	25.1	26.6	26.6
	10% MCWB	23.2	23.6	22.9	21.1	18.9	16.4	16.2	17.5	18.7	19.8	21.2	22.4	22.4
	WB	25.9	26.1	25.5	24.0	22.0	19.8	19.1	20.3	21.6	22.6	24.1	25.4	25.4
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	29.4	29.6	28.6	27.1	25.1	23.5	22.2	23.7	25.5	26.6	27.3	29.0	29.0
	2% WB	25.1	25.4	24.7	23.0	21.1	18.8	18.3	19.5	20.6	21.6	23.3	24.5	24.5
	2% MCDB	28.2	28.6	27.8	26.3	24.2	22.5	22.0	22.8	24.0	25.0	26.2	27.6	27.6
	5% WB	24.5	24.9	24.2	22.3	20.4	18.0	17.6	18.9	20.0	20.9	22.6	23.7	23.7
	5% MCDB	27.5	27.8	27.3	25.6	23.8	22.0	21.5	22.3	23.0	23.9	25.3	26.5	26.5
	10% WB	23.8	24.4	23.6	21.7	19.7	17.2	16.9	18.3	19.5	20.4	21.8	23.1	23.1
	10% MCDB	26.8	27.3	26.7	25.0	23.2	21.2	20.9	21.7	22.5	23.2	24.4	25.9	25.9
(35) Mean Daily Temperature Range	MDBR	6.0	6.3	6.6	7.6	9.5	11.1	10.6	9.0	7.4	6.3	6.1	6.0	6.0
	5% DB	7.2	7.3	7.6	9.1	11.1	13.5	13.1	11.5	9.1	8.2	7.7	7.4	7.4
	5% MCWBR	3.8	3.7	4.0	4.8	6.5	8.0	8.0	6.7	5.3	4.6	4.4	4.1	4.1
	5% WB	6.7	6.7	7.1	8.2	9.7	11.4	11.5	9.8	8.1	7.2	7.0	6.8	6.8
	5% MCWBR	3.9	3.7	4.1	4.7	6.1	7.8	7.8	6.3	4.9	4.3	4.2	4.1	4.1
(40) Clear-Sky Solar Irradiance	taub	0.456	0.447	0.438	0.407	0.381	0.365	0.375	0.438	0.487	0.496	0.470	0.463	0.463
	taud	2.301	2.333	2.339	2.389	2.424	2.446	2.406	2.218	2.099	2.116	2.228	2.267	2.267
	Ebn at Noon	889	879	852	826	799	790	796	780	792	825	871	886	886
	Edh at Noon	140	132	124	107	93	86	93	124	155	162	150	146	146
(44) All-Sky Solar Radiation	RadAvg	5.91	5.78	5.02	4.09	3.48	3.10	3.31	3.86	4.45	4.88	5.39	5.80	5.80
(45)	RadStd	0.47	0.43	0.24	0.26	0.18	0.15	0.22	0.23	0.28	0.45	0.43	0.43	0.43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page